8.1 The next step is to develop block diagram(s) showing the functional flow sequence of the system, both for technical understanding of the functions and operation of the system, and for the subsequent analysis. As a minimum the block diagram should contain:
.1 breakdown of the system into major sub-systems or equipment:
.2 all appropriate labelled inputs and outputs and identification numbers by which each sub-system is consistently referenced: and
.3 all redundancies, alternative signal paths and other engineering features which provide "fail-safe" measures.
An example of a system block diagram is given at appendix 1.
8.2 It may be necessary to have a different set of block diagrams prepared for each operational mode.
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features which provide "fail-safe" measures.
An example of a system block diagram is given at appendix 1.
8.2 It may be necessary to have a different set of block diagrams prepared for each operational mode.
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ADOPTION OF THE INTERNATIONAL CODE OF SAFETY
FOR HIGH SPEED CRAFT 9 Identification of failure modes, causes and effects
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9.1 Failure mode is the manner by which a failure is observed. It generally describes the way the failure occurs and its impact on the equipment or . system. As an example, a list of failure modes is given in table 1. The failure modes listed in table 1 can describe the failure of any system element in sufficiently specific terms. When used in conjunction with performance specifications governing the inputs and outputs on the system block diagram, all potential failure modes can be thus identified and described. Thus, for example, a power supply may have a failure mode described as "loss of output"(29), and a failure cause "open (electrical)"(3l).
9.2 A failure mode in a system element could also be the failure cause of a

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example, a power supply may have a failure mode described as "loss of output"(29), and a failure cause "open (electrical)"(3l).
9.2 A failure mode in a system element could also be the failure cause of a system failure. For example, the hydraulic line of a steering gear system might have a failure mode of "external leakage"(lO). This failure mode of the hydraulic line could become a failure cause of the steering gear system's failure mode "loss of output"(29).
9.3 Each system should be considered in a top-down approach, starting from the system's functional output, and failure should be assumed by one possible cause at a time. Since a failure mode may have more than one cause, all potential independent causes for each failure mode should be identified.

9.4 If major systems can fail without any adverse effect there is no need to consider them further unless the failure can go undetected by an operator. To decide that there is no adverse effect does not mean just the identification of system redundancy. The redundancy should be shown to be immediately effective or brought on line with negligible time lag. In addition, if the sequence is:

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decide that there is no adverse effect does not mean just the identification of system redundancy. The redundancy should be shown to be immediately effective or brought on line with negligible time lag. In addition, if the sequence is:
"failure-alarm-operator action - start of back up - back up in service",
the effects of delay should be considered.
10 Failure effects
10.1 The consequence of a failure mode on the operation, function, or status of an equipment or a system is called a "failure effect". Failure effects on a specific sub-system or equipment under consideration are called "local failure effects". The evaluation of local failure effects will help to determine the effectiveness of any redundant equipment or corrective action at that system level. In certain instances, there may not be a local effect beyond the failure mode itself.
10.2 The impact of an equipment or sub-system failure on the system output (system function) is called an "end effect". End effects should be evaluated and their severity classified in accordance with the following categories:
.1 catastrophic;
.2 hazardous;
.3 major; and
.4 minor.
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.1 catastrophic;
.2 hazardous;
.3 major; and
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The definitions of these four categories of failure effects are given in 2.3 of annex 3 of this Code.
10.3 If the end effect of a failure is classified as hazardous or catastrophic, back-up equipment is usually required to prevent or minimize such effect. For hazardous failure effects corrective operational procedures

may be accepted.
11 Failure detection
11.1 The FMEA study in general only analyses failure effects based on a single failure in the system and therefore a failure detection means, such as visual or audible warning devices, automatic sensing devices, sensing instrumentation or other unique indications should be identified.
11.2 Where the system element failure is non-detectable (i.e. a hidden fault or any failure which does not give any visual or audible indication to the operator) and the system can continue with its specific operation, the

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11.2 Where the system element failure is non-detectable (i.e. a hidden fault or any failure which does not give any visual or audible indication to the operator) and the system can continue with its specific operation, the analysis should be extended to determine the effects of a second failure, which in combination with the first undetectable failure may result in a more severe failure effect, e.g., hazardous or catastrophic effect.
12 Corrective measures
12.1 The response of any back-up equipment, or any corrective action initiated at a given system level to prevent or reduce the effect of the failure mode of a system element or equipment, should also be identified and evaluated.
12.2 Provisions which are features of the design at any system level to nullify the effects of a malfunction or failure, such as controlling or deactivating system elements to halt generation or propagation of failure effects, or activating back-up or standby items or systems, should be described. Corrective design provisions include:
.1 redundancies that allow continued and safe operation;
.2 safety devices, monitoring or alarm provisions, which permit

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effects, or activating back-up or standby items or systems, should be described. Corrective design provisions include:
.1 redundancies that allow continued and safe operation;
.2 safety devices, monitoring or alarm provisions, which permit restricted operation or limit damage; and
.3 alternative modes of operation.
12.3 Provisions which require operator action to circumvent or mitigate the effects of the postulated failure should be described. The possibility and effect of operator error should be considered, if the corrective action or the initiation of the redundancy requires operator input, when evaluating the means to eliminate the local failure effects.
12.4 It should be noted that corrective responses acceptable in one operational mode may not be acceptable at another, e.g., a redundant system element with considerable time lag to be brought into line, while meeting the operational mode "normal seagoing conditions at full speed" may result in a catastrophic effect in another operational mode, e.g., "maximum permitted operating speed in congested water".
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catastrophic effect in another operational mode, e.g., "maximum permitted operating speed in congested water".
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FOR HIGH SPEED CRAFT 13 Use of probability concept
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13.1 If corrective measures or redundancy as described in preceding paragraphs are not provided for any failure, as an alternative the probability of occurrence of such failure should meet the following criteria of acceptance:
.1 a failure mode which results in a catastrophic effect should be assessed to be extremely improbable;
.2 a failure mode assessed as extremely remote should not result in worse than hazardous effects; and
.3 a failure mode assessed as either frequent or reasonably probable